AI for Medical Imaging

Master Artificial Intelligence

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Fall 2026

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2026/09/29 – Reconstruction, denoising & Foundational models

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Announcements

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Exam preparation: a bit at the end of the next two lectures

Next week: guest lectures. Content still at the exam

Currently happening: MICCAI 2026 in Strasbourg
https://conferences.miccai.org/2026/en/default.asp

Many workshops, among them: Open data initiative:
https://doi.org/10.59275/j.melba.2026-6e9a

And MICCAI 2030 will be in...

2026/09/29 – Reconstruction, denoising & Foundational models

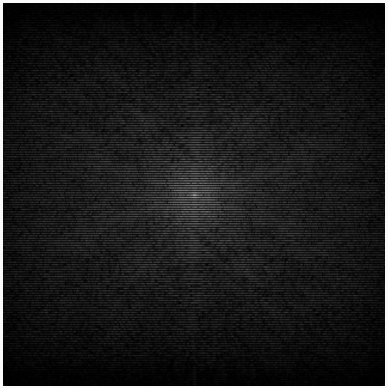
Reconstruction

Motivation: reminder of MRI acquisition

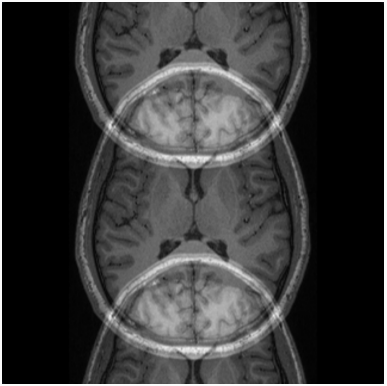
1. Align all hydrogens atoms with the magnet
2. Emit a radiofrequency wave
3. Measure how quickly the atoms go back into alignment
4. Repeat for several frequencies
5. Reconstruct the image

https://youtu.be/1CGzk-nV06g, extra read:
https://nationalmaglab.org/magnet-academy/read-science-stories/science-simplified/mri-a-guided-tour/

Under-sampling the k-space



Under-sampled (factor 2) k-space

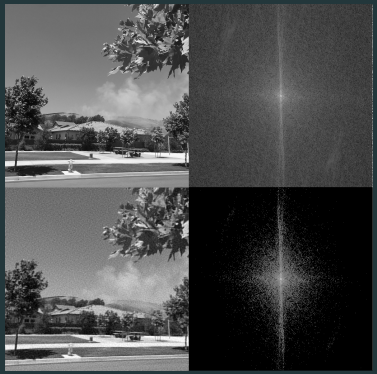


Reconstructed MRI image

https://kspace-explorer.streamlit.app/

Relation with JPEG and image compression

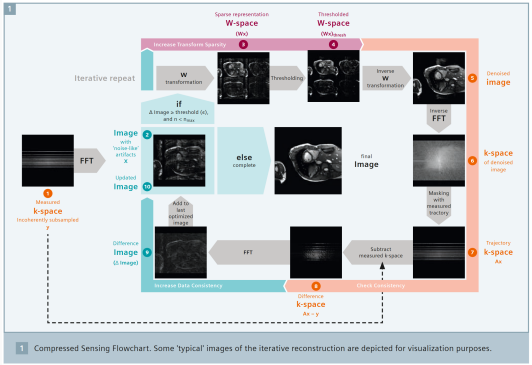
Similar idea: remove high-frequencies to "compress" the image



From https://blog.demofox.org/2020/11/04/frequency-domain-image-compression-and-filtering/

Accelerated MRI reconstruction: compressed sensing

Ill-posed problems: many possible solutions. Iterative algorithms to tackle it



1 Compressed Sensing Flowchart. Some 'typical' images of the iterative reconstruction are depicted for visualization purposes.

https://www.siemens-healthineers.com/magnetic-resonance-imaging/clinical-specialities/compressed-sensing

Learning MRI reconstruction from sub-sampled spaces (i)

Ω: image space, Ξ: spectral domain

$\mu \in \mathbb{C}^{\Xi}$: k-space information, $x \in \mathbb{R}^{\Omega}$

$\mathcal{F} : \Xi \to \Omega$ Fourier transform

$\mathcal{F}^{-1} : \Omega \to \Xi$ Inverse Fourier transform

$\mathcal{D} = \{x_n\}_{n=1}^{N}$: dataset of fully-sampled images

$\mu_n = \mathcal{F}^{-1}(x_n)$

Learning MRI reconstruction from sub-sampled spaces (ii)

$\tilde{\mu}_n$: artificially sub-sampled k-space

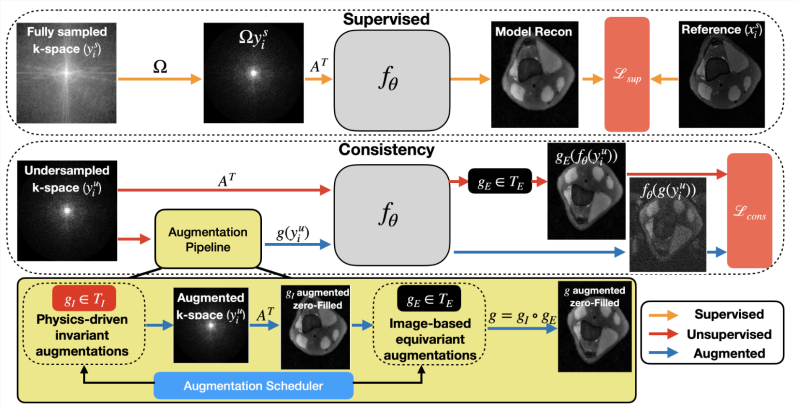
$\mathcal{N}_\theta : \tilde{\mu}_n \mapsto \tilde{x}_{\theta,n}$: network with param. θ reconstructing an image.

$$\min_\theta \mathcal{L}_{\text{rec}}(\tilde{x}_\theta, x)$$

For instance:

$$\mathcal{L}_{\text{rec}}(\tilde{x}_\theta, x) \propto \sum_{i \in \Omega} \left(\tilde{x}_\theta^{(i)} - x^{(i)} \right)^2$$

Adding self-supervision (i)



VORTEX: Physics-Driven Data Augmentations Using Consistency Training for Robust Accelerated MRI Reconstruction

Arjun D Desai, et al., MIDL 2022, `https://2022.midl.io/papers/b_l_3`, Best paper award

Adding self-supervision (ii)

Augmentation strategy $A : \tilde{\mu} \mapsto \tilde{\tilde{\mu}}$

$\tilde{\tilde{x}}_\theta := \mathcal{N}_\theta(\tilde{\tilde{\mu}})$

Consistency loss, for instance:

$$\mathcal{L}_{\text{cons.}}(\tilde{x}_\theta, \tilde{\tilde{x}}_\theta) \propto \sum_{i \in \Omega} \left(\tilde{x}_\theta^{(i)} - \tilde{\tilde{x}}_\theta^{(i)} \right)^2$$

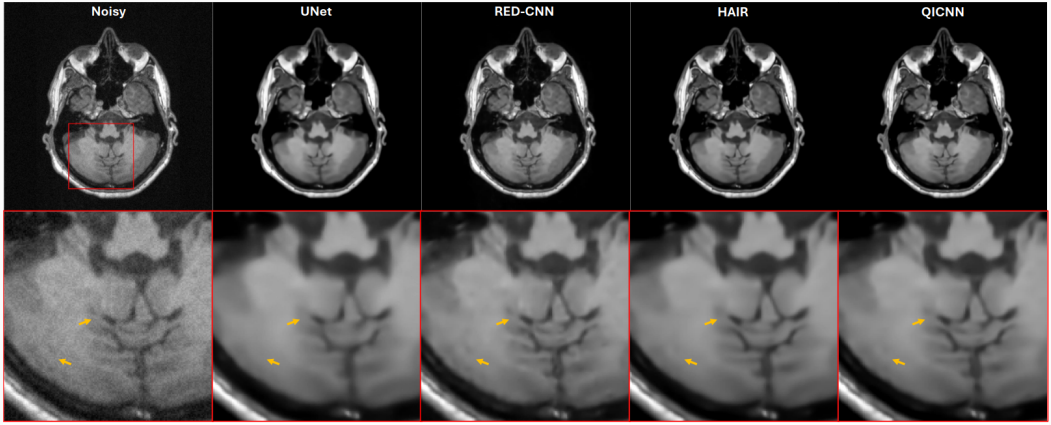
$$\min_\theta \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{cons.}}$$

$\lambda \in \mathbb{R}$

2026/09/29 – Reconstruction, denoising & Foundational models

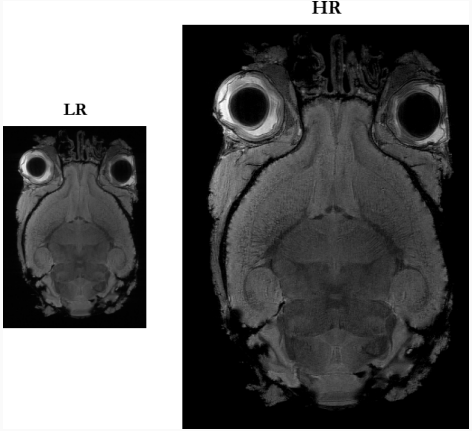
"Generative imaging": Denoising, super-resolution, modality generation, ...

Denoising



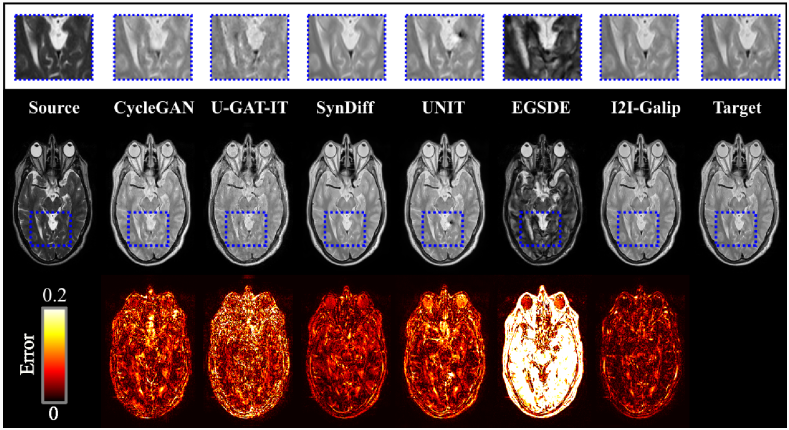
From *Quantum-Inspired Orthonormal CNN for Energy-Efficient Medical Image Denoising*, Sayantan Dutta, MIDL 2026, https://proceedings.mlr.press/v315/kumar26a.html

Super-resolution



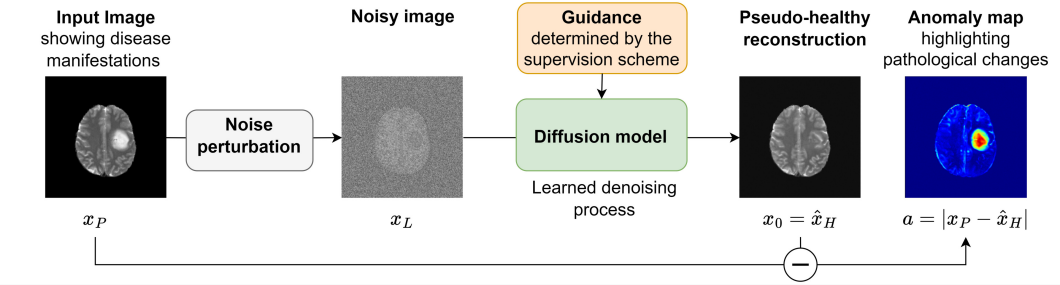
From *SRMRI: An Open Dataset and Benchmarks for Blind Super-Resolution of 2D MRI*, A. Poudel, M. Shrestha, N. Wang, U. Nakarmi, MIDL 2026, https://openreview.net/pdf?id=25wvW3T7AR

Modality generation/image translation



From *I2I-Galip: Unsupervised Medical Image Translation Using Generative Adversarial CLIP*, Y. Korkmaz, V. Patel, MIDL 2026, https://openreview.net/pdf?id=lAQ29DUZCa

Other applications: denoising for anomaly detection



Denoising Diffusion Models for Anomaly Localization in Medical Images, C. Bercea, P. Cattin, J. Schnabel, J. Wolleb, Melba journal, 2025, https://doi.org/10.59275/j.melba.2025-c586

Similarities, and limits?

Similar idea for all tasks: “learn” to generate, recover, the missing information

Limits? Issues?

What are we measuring? Do we have the information? Different modalities exists for a reason.

2026/09/29 – Reconstruction, denoising & Foundational models

Foundational models in medical imaging

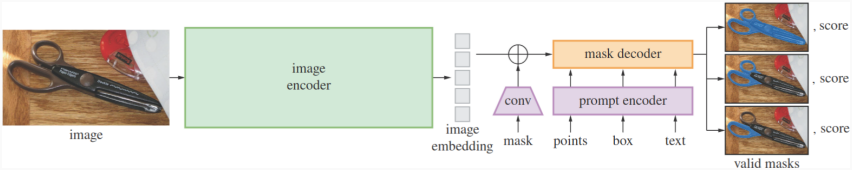
Foundational models

Any general model trained on a “big enough” dataset (popular word to throw in).

Distinguish different types here:

- promptable (or not) models;
- general or medical models;
- fined-tuned or not models;
- supervised vs self-supervised models.

SAM: Segment Anything

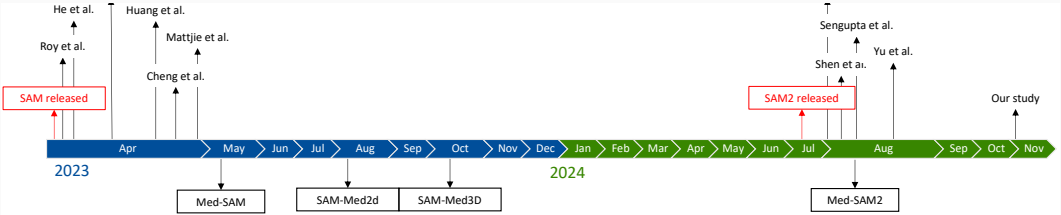


Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., ... & Girshick, R. (2023, October). Segment anything. In 2023 IEEE/CVF international conference on computer vision (ICCV) (pp. 3992-4003). IEEE.

Developped and trained on 2D natural images. SAM2, SAM3: more powerful, also more costly to run.

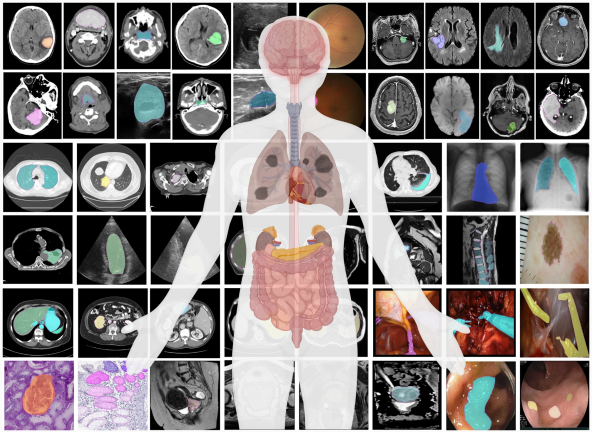
https://segment-anything.metademolab.com/

Fine-tuning SAM for medical applications (i)



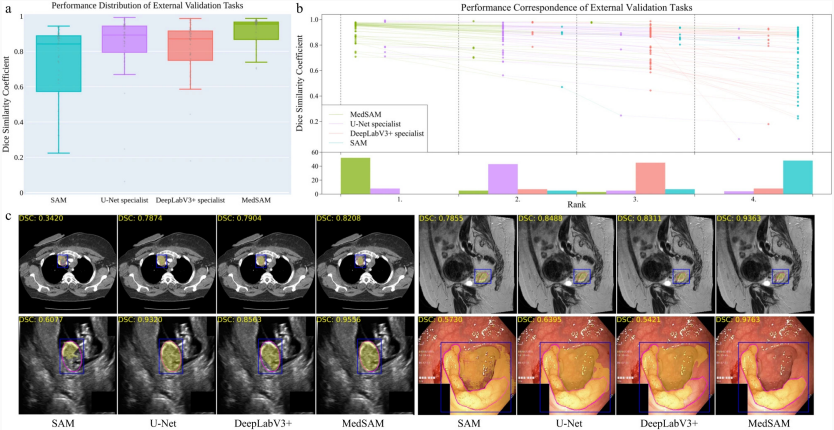
Older timeline of, from Magg, C., Kervadec, H., & Sánchez, C. I. (2024). *Zero-shot capability of SAM-family models for bone segmentation in CT scans*. arXiv preprint arXiv:2411.08629.

Fine-tuning SAM for medical applications (ii)



Ma, J., He, Y., Li, F., Han, L., You, C., & Wang, B. (2024). *Segment anything in medical images.* Nature communications, 15(1), 654.

Fine-tuning SAM for medical applications (iii)



Ma, J., He, Y., Li, F., Han, L., You, C., & Wang, B. (2024). *Segment anything in medical images.* Nature communications, 15(1), 654.

Special case: 3D MRI as video

SAM and derivative architectures: limited to 2D.

SAM2 and more: video mode, used for 3D scans

Domain shift?

Training set collected on domain D_1:

$$\mathcal{D} = \{x_n\}_{n=1}^{N}, \quad x_n \sim D_1 \quad \forall n$$

Second domain D_2 at inference, $x_m \sim D_2$

This is referred as *domain shift*, or sometimes *data drift*.

One of the main cause of poor generalization

Sources of domain shift

Different scanner

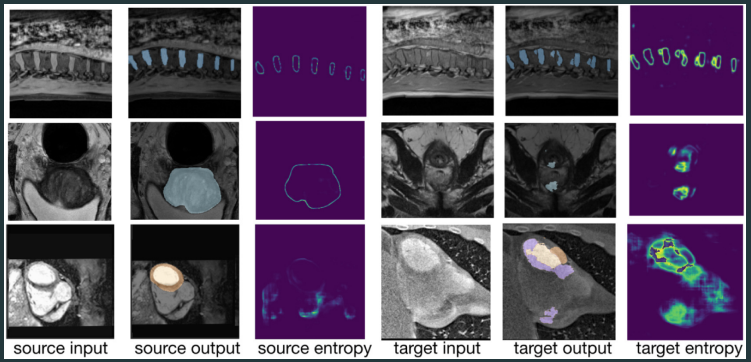
Different scanner vendor

Different scanning protocol

Image is operator dependent

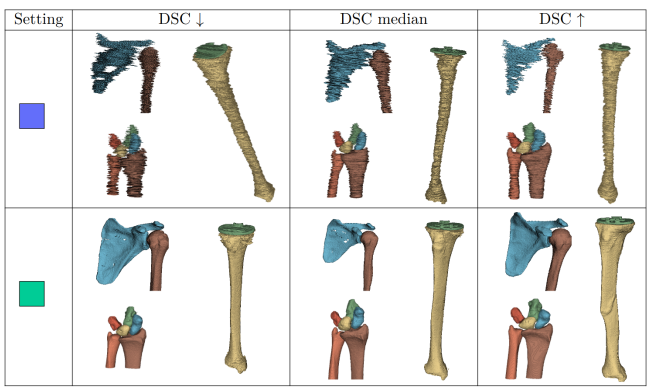
Different populations: examples?

Domain adaptation: from *source* to *target* domains



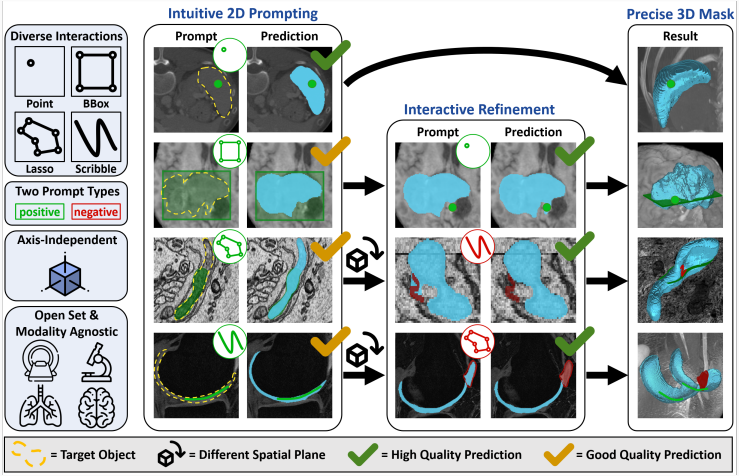
Top: 2 spine images from Water (left) and In-Phase (right) MRI [...]. Middle: 2 prostate MRI images from different sites. Bottom: 2 cardiac images from MRI (left) and CT (right). [...] The domain shift in the target causes a drop in confidence and accuracy. From: *Source-free domain adaptation for image segmentation*. Bateson, M., Kervadec, H., Dolz, J., Lombaert, H., & Ayed, I. B. (2022). Medical Image Analysis, 82, 102617.

Trade-off between in-domain performances and good generalization



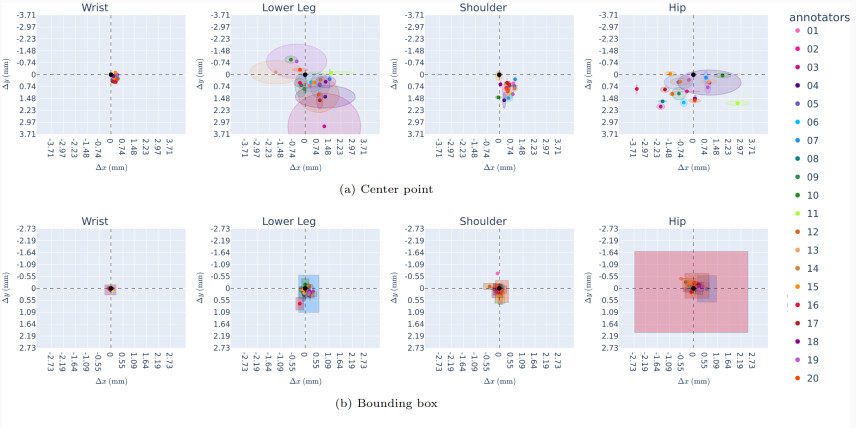
Performances on a private dataset (Amsterdam UMC). From C. Magg et al. *Zero-shot capability of SAM-family models for bone segmentation in CT scans*. arXiv preprint arXiv:2411.08629. (2024)

nnInteractive



Same team as nnU-Net: https://github.com/mic-dkfz/nninteractive

Stability in the face of user variability (i)

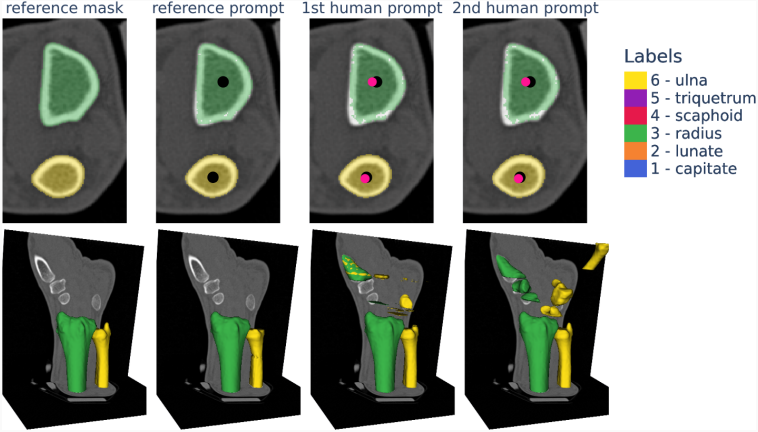


(a) Center point

(b) Bounding box

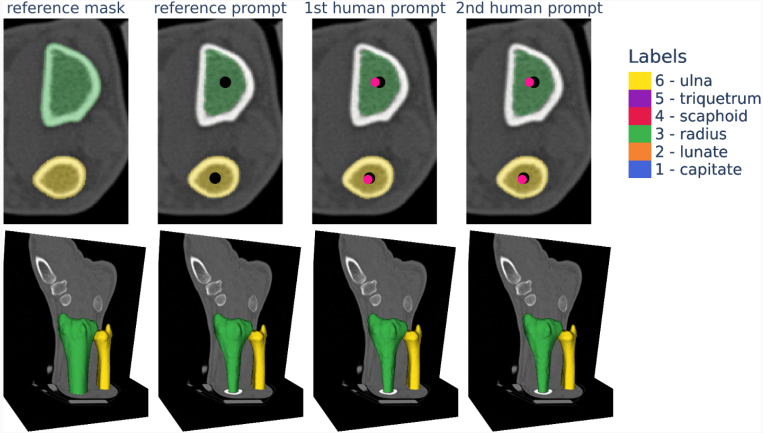
Spatial distribution of prompt placement per annotator per dataset. From Magg, C. et al. (2026). *Prompting with the human-touch: evaluating model-sensitivity of foundation models for musculoskeletal CT segmentation*. arXiv preprint arXiv:2603.10541.

Stability in the face of user variability (ii)



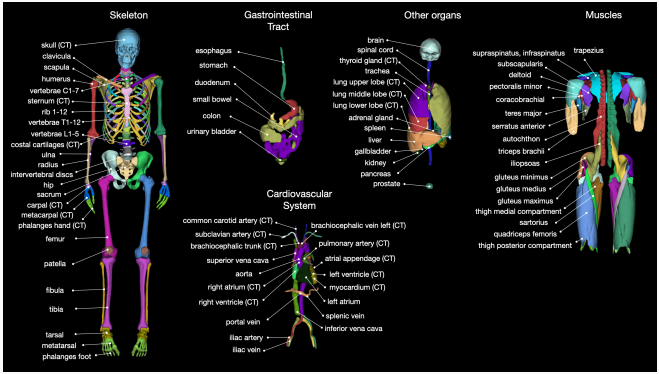
Variation of segmentation for SAM 2.1T for slightly different points

Stability in the face of user variability (iii)



Variation of segmentation for nnInteractive for slightly different points

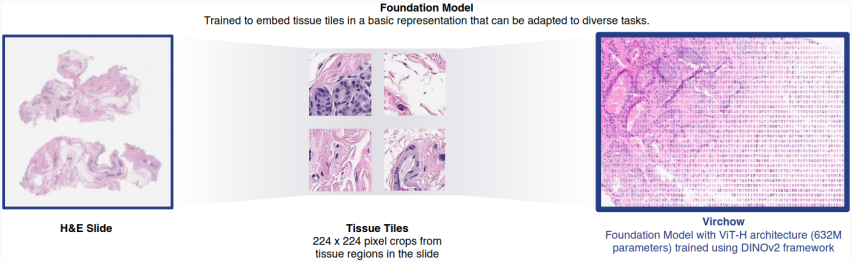
TotalSegmentator: a dataset, and a trained model



https://github.com/wasserth/TotalSegmentator

1939 CTs: https://zenodo.org/records/22688904, 1296 MRIs: https://zenodo.org/records/22688334

Foundational models for embeddings: one example



Vorontsov, Eugene, et al. "*Virchow: A million-slide digital pathology foundation model.*" arXiv preprint arXiv:2309.07778 (2023).

Trained 1.5 millions H&E slides using the DINOv2 algorithm.

2026/09/29 – Reconstruction, denoising & Foundational models

Summary

title

- *Learned* image reconstruction
- and related applications
- Domain shift and domain adaptation to not be under-estimated
- Foundational models

Next time: classification (i.e., decision/prediction).